

b.1

树:



二叉树



b.1b 按书上结构体定义方式.

```
void swap (BiTree T) {
    if (!T) return;
    BiTree temp = T->lchild;
    T->lchild = T->rchild;
    T->rchild = temp;
    swap (T->lchild);
    swap (T->rchild);
}
```

b.2 2个

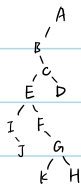
b.3 (1) 所有结点左子树为空 / 空二叉树

(2) 所有结点右子树为空 / 空二叉树

(3) 只有根结点 / 空二叉树

b.4 (1) 第h层: k^{h-1} (2) $\lfloor \frac{i-1}{k} \rfloor + 1$ (3) $(i-1)k + 1 + j$ (4) $(i-1) \bmod k \neq 0, i+1$

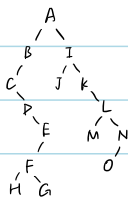
b.8



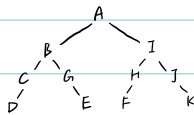
先根遍历: ABCDEIJFGKHD

后根遍历: BIJEFKGHCDA

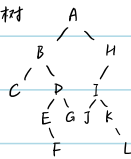
b.9



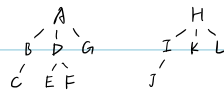
b.11



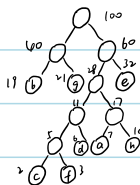
b.13 对应二叉树



森林



b.14 (1)



(2)

a	1010
b	00
c	10000
d	1001
e	11
f	10001
g	01
h	1011

(3) WPL = $7 \times 4 + 19 \times 2 + 2 \times 5 + 6 \times 4 + 3 \times 2 + 3 \times 5 + 21 \times 2 + 10 \times 4 = 261$

b.22

```
int TreeDepth (CSTree T) {
    int maxh = 0;
    if (!T) return 0;
    CSTree p; int h;
    for (p = T->firstchild; p; p = p->nextsibling) {
        h = TreeDepth (p);
        if (h > maxh) maxh = h;
    }
    return maxh + 1;
}
```

b.23

```
int Count (BiTree T, int k) {
    if (!root) return 0;
    if (k < 1) return 0;
    if (k == 1) return 1;
    return Count (T->lchild, k-1) + Count (T->rchild, k-1);
}
```